



Examples 2.28 – 2.36

Complex Numbers — De Moivre's Theorem & Roots of Unity

Detailed, step-by-step solutions to all 9 worked examples



2.28

If $z = \cos\theta + i\sin\theta$, show $z^n + 1/z^n = 2\cos n\theta$ and $z^n - 1/z^n = 2i \sin n\theta$

Step 1: Apply De Moivre's theorem to z^n .

$$z = \cos\theta + i\sin\theta \implies z^n = (\cos\theta + i\sin\theta)^n = \cos n\theta + i \sin n\theta$$

Step 2: Find $1/z^n$ using the same theorem.

$$1/z^n = (\cos\theta + i\sin\theta)^{-n} = \cos n\theta - i \sin n\theta$$

Step 3: Add and subtract the two results.

$$z^n + 1/z^n = (\cos n\theta + i\sin n\theta) + (\cos n\theta - i\sin n\theta) = 2\cos n\theta$$

$$z^n - 1/z^n = (\cos n\theta + i\sin n\theta) - (\cos n\theta - i\sin n\theta) = 2i \sin n\theta$$

Hence $z^n + 1/z^n = 2\cos n\theta$ and $z^n - 1/z^n = 2i \sin n\theta$ — proved.

2.29

Simplify $(\sin \pi/6 + i \cos \pi/6)^{18}$

Step 1: Rewrite $\sin\theta + i\cos\theta$ using $i \cdot \text{cis}(-\theta)$.

$$i \cdot \text{cis}(-\theta) = i(\cos\theta - i\sin\theta) = i\cos\theta + \sin\theta = \sin\theta + i\cos\theta$$

$$\text{So } \sin(\pi/6) + i\cos(\pi/6) = i \cdot \text{cis}(-\pi/6)$$

Step 2: Raise both sides to the power 18.

$$[\sin(\pi/6) + i\cos(\pi/6)]^{18} = i^{18} \cdot \text{cis}(-18\pi/6) = i^{18} \cdot \text{cis}(-3\pi)$$

Step 3: Simplify i^{18} and $\text{cis}(-3\pi)$.

$$i^{18} = (i^4)^4 \cdot i^2 = 1 \cdot (-1) = -1$$

$$\text{cis}(-3\pi) = \cos(3\pi) - i \sin(3\pi) = -1 - 0 = -1$$

$$\text{Result} = (-1) \times (-1) = 1$$

2.30

Simplify $[(1+\cos 2\theta + i\sin 2\theta) / (1+\cos 2\theta - i\sin 2\theta)]^{30}$

Step 1: Let $z = \cos 2\theta + i\sin 2\theta$. Since $|z| = 1$, we get $1/z = \bar{z} = \cos 2\theta - i\sin 2\theta$.

Step 2: Rewrite numerator and denominator in terms of z .

Numerator: $1 + \cos 2\theta + i\sin 2\theta = 1 + z$

Denominator: $1 + \cos 2\theta - i\sin 2\theta = 1 + 1/z = (z+1)/z$

Step 3: Divide.

$$(1+z) \div [(1+z)/z] = z$$

Step 4: Raise to the power 30 and apply De Moivre's theorem.

$$z^{30} = (\cos 2\theta + i\sin 2\theta)^{30} = \cos 60\theta + i\sin 60\theta$$

Result = $\cos 60\theta + i \sin 60\theta$

2.31 (i) Simplify $(1 + i)^{18}$

Step 1: Convert $1+i$ to polar form.

$$r = \sqrt{1^2+1^2} = \sqrt{2}, \quad \theta = \tan^{-1}(1/1) = \pi/4 \text{ (first quadrant)}$$

$$1 + i = \sqrt{2} (\cos \pi/4 + i \sin \pi/4)$$

Step 2: Raise to power 18 using De Moivre's theorem.

$$(1+i)^{18} = (\sqrt{2})^{18} [\cos(18 \cdot \pi/4) + i \sin(18 \cdot \pi/4)] = 2^9 [\cos(9\pi/2) + i \sin(9\pi/2)]$$

Step 3: Reduce the angle.

$$9\pi/2 = 4\pi + \pi/2 \implies \cos(9\pi/2) + i \sin(9\pi/2) = \cos(\pi/2) + i \sin(\pi/2) = i$$

$$(1+i)^{18} = 2^9 \cdot i = 512i$$

2.31 (ii) Simplify $(-\sqrt{3} + 3i)^{31}$

Step 1: Convert $-\sqrt{3}+3i$ to polar form.

$$r = \sqrt{(-\sqrt{3})^2 + 3^2} = \sqrt{12} = 2\sqrt{3}$$

Reference angle = $\tan^{-1}(3/\sqrt{3}) = \tan^{-1}(\sqrt{3}) = \pi/3$; point lies in Quadrant II

$$\theta = \pi - \pi/3 = 2\pi/3$$

$$-\sqrt{3}+3i = 2\sqrt{3} (\cos 2\pi/3 + i \sin 2\pi/3)$$

Step 2: Raise to power 31 and reduce the angle.

$$(-\sqrt{3}+3i)^{31} = (2\sqrt{3})^{31} [\cos(62\pi/3) + i \sin(62\pi/3)]$$

$$62\pi/3 = 20\pi + 2\pi/3 \Rightarrow \text{angle reduces to } 2\pi/3$$

Step 3: Substitute $\cos(2\pi/3) = -1/2$, $\sin(2\pi/3) = \sqrt{3}/2$, and $(2\sqrt{3})^{31} = 2^{31} \cdot 3^{15} \sqrt{3}$.

$$\text{Result} = 2^{31} \cdot 3^{15} \sqrt{3} (-1/2 + i\sqrt{3}/2) = -2^{30} \cdot 3^{15} \sqrt{3} + i \cdot 2^{30} \cdot 3^{16}$$

2.32

Find the cube roots of unity

Step 1: We need $z = 1^{1/3}$, i.e. solve $z^3 = 1$.

In polar form: $z^3 = \cos(2k\pi) + i\sin(2k\pi)$, $k = 0, 1, 2$

$\Rightarrow z = \cos(2k\pi/3) + i\sin(2k\pi/3)$, $k = 0, 1, 2$

Step 2: Substitute $k = 0, 1, 2$.

$k=0$: $z = \cos 0 + i\sin 0 = 1$

$k=1$: $z = \cos(2\pi/3) + i\sin(2\pi/3) = -1/2 + i\sqrt{3}/2$

$k=2$: $z = \cos(4\pi/3) + i\sin(4\pi/3) = -1/2 - i\sqrt{3}/2$

Cube roots of unity: $1, (-1+i\sqrt{3})/2, (-1-i\sqrt{3})/2$ i.e. $1, \omega, \omega^2$

2.33

Find the fourth roots of unity

Step 1: We need $z = 1^{1/4}$, i.e. solve $z^4 = 1$.

In polar form: $z^4 = \cos(2k\pi) + i\sin(2k\pi)$, $k = 0, 1, 2, 3$

$\Rightarrow z = \cos(2k\pi/4) + i\sin(2k\pi/4) = \cos(k\pi/2) + i\sin(k\pi/2)$, $k = 0, 1, 2, 3$

Step 2: Substitute $k = 0, 1, 2, 3$.

$k=0$: $z = \cos 0 + i\sin 0 = 1$

$k=1$: $z = \cos(\pi/2) + i\sin(\pi/2) = i$

$k=2$: $z = \cos \pi + i\sin \pi = -1$

$k=3$: $z = \cos(3\pi/2) + i\sin(3\pi/2) = -i$

Fourth roots of unity: $1, i, -1, -i$ i.e. $1, \omega, \omega^2, \omega^3$

2.34

Solve $z^3 + 8i = 0$, $z \in \mathbb{C}$

Step 1: Write $z^3 = -8i$ and convert $-8i$ to polar form.

$$|-8i| = 8, \text{ argument} = -\pi/2 \Rightarrow z^3 = 8[\cos(-\pi/2 + 2k\pi) + i\sin(-\pi/2 + 2k\pi)], k=0,1,2$$

Step 2: Take the cube root.

$$z = 2[\cos((- \pi/2 + 2k\pi)/3) + i\sin((- \pi/2 + 2k\pi)/3)], k = 0, 1, 2$$

Step 3: Substitute $k = 0, 1, 2$.

$$k=0: \text{ angle} = -\pi/6 \Rightarrow z = 2(\cos(-30^\circ) + i\sin(-30^\circ)) = \sqrt{3} - i$$

$$k=1: \text{ angle} = \pi/2 \Rightarrow z = 2(\cos 90^\circ + i\sin 90^\circ) = 2i$$

$$k=2: \text{ angle} = 7\pi/6 \Rightarrow z = 2(\cos 210^\circ + i\sin 210^\circ) = -\sqrt{3} - i$$

Roots: $z = \sqrt{3} - i, 2i, -\sqrt{3} - i$

2.35

Find all cube roots of $\sqrt{3} + i$

Step 1: Convert $\sqrt{3}+i$ to polar form.

$$r = \sqrt{3+1} = 2, \quad \theta = \tan^{-1}(1/\sqrt{3}) = \pi/6 \implies \sqrt{3}+i = 2(\cos \pi/6 + i \sin \pi/6)$$

Step 2: Take the cube root using De Moivre's formula.

$$z = 2^{1/3} [\cos((\pi/6+2k\pi)/3) + i \sin((\pi/6+2k\pi)/3)], \quad k = 0, 1, 2$$

Step 3: Substitute $k = 0, 1, 2$.

$$k=0: \quad z = 2^{1/3} [\cos(\pi/18) + i \sin(\pi/18)]$$

$$k=1: \quad z = 2^{1/3} [\cos(13\pi/18) + i \sin(13\pi/18)]$$

$$k=2: \quad z = 2^{1/3} [\cos(25\pi/18) + i \sin(25\pi/18)] = 2^{1/3} [-\cos(7\pi/18) - i \sin(7\pi/18)]$$

2.36

Equilateral triangle inscribed in $|z|=2$, $z_1=1+i\sqrt{3}$ — find z_2, z_3

Setup: $|z| = 2$ is a circle, centre origin, radius 2. For an equilateral triangle inscribed in it, each vertex is obtained by rotating the previous one by $2\pi/3$ (120°) about the origin.
(Multiplying z by $e^{i\theta}$ rotates z by θ radians anticlockwise about the origin.)

Step 1: Find z_2 by rotating z_1 by $2\pi/3$.

$$\begin{aligned} z_2 &= z_1 \cdot \text{cis}(2\pi/3) = (1+i\sqrt{3})(-1/2+i\sqrt{3}/2) = -1/2+i\sqrt{3}/2-i\sqrt{3}/2+i^2(3/2) \\ &= -1/2 - 3/2 = -2 \end{aligned}$$

Step 2: Find z_3 by rotating z_1 by $4\pi/3$ (or z_2 by $2\pi/3$).

$$z_3 = z_2 \cdot \text{cis}(2\pi/3) = -2(-1/2+i\sqrt{3}/2) = 1 - i\sqrt{3}$$

$$z_2 = -2 \quad \text{and} \quad z_3 = 1 - i\sqrt{3}$$

Thank You

Examples 2.28 – 2.36 — Complex Numbers, Solutions Complete

